

Tutorial 6 – Week 6

Midterm Review

MTE 252

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Question 2.2-3

2.2-3 A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^3 + 9D) \{y(t)\} = (2D^3 + 1) \{x(t)\}.$$

- (a) What is the characteristic equation of this system?
- (b) What are the characteristic modes of this system?
- (c) Assuming $y_{\text{zir}}(0) = 4$, $\dot{y}_{\text{zir}}(0) = -18$, and $\ddot{y}_{\text{zir}}(0) = 0$, determine this system's zero-input response $y_{\text{zir}}(t)$. Simplify $y_{\text{zir}}(t)$ to include only real terms (i.e., no j 's should appear in your answer).



Solution 2.2-3

Solution 2.2-3

Question 2.5-1

2.5-1 Explain, with reasons, whether the LTIC systems described by the following equations are (i) stable or unstable in the BIBO sense; (ii) asymptotically stable, unstable, or marginally stable. Assume that the systems are controllable and observable.

(b) $D(D^2 + 3D + 2)y(t) = (D + 5)x(t)$

(d) $(D + 1)(D^2 - 6D + 5)y(t) = (3D + 1)x(t)$

2.5-2 Repeat Prob. 2.5-1 for the following:

(c) $(D + 1)(D^2 + 9)^2y(t) = (2D + 9)x(t)$

Solution 2.5-1

Question 2.4-5

2.4-5 Suppose an LTIC system has impulse response $h(t)$ and input $x(t) = u(t)$. Figure P2.4-5 shows $x(t)$ and $h(t + 1)$, respectively. Be careful! Figure P2.4-5 shows $h(t + 1)$, not $h(t)$.

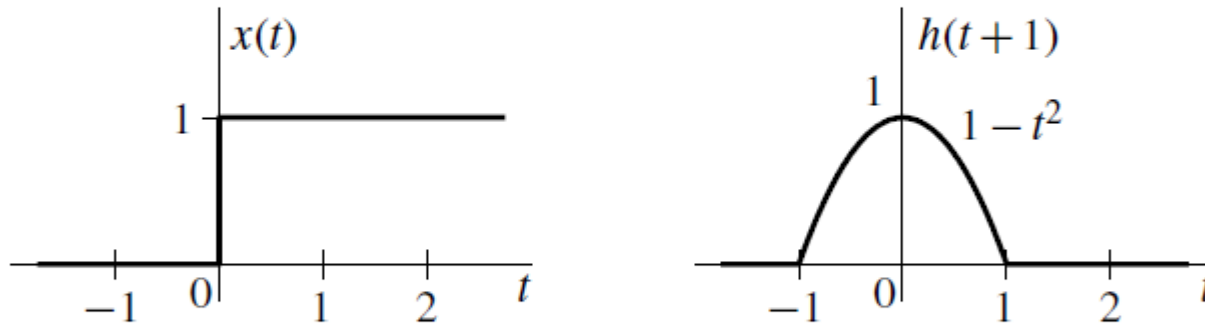
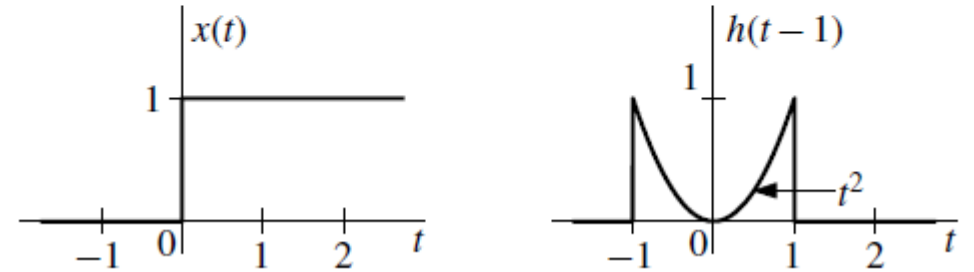


Figure P2.4-5

2.4-6 Repeat Prob. 2.4-5 using the signals of Fig. P2.4-6, rather than those of Fig. P2.4-5. Be careful! Figure P2.4-6 shows $h(t - 1)$, not $h(t)$.



- (a) Is system $h(t)$ causal? Mathematically justify your answer.

Solution 2.4-5 and 6